

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

Name Riya Verma

Enrollment No. 23805022

Jaypee Institute of Information Technology, Noida

T1 Examination, Even 2024

B.Tech/Dual (CSE/IT/ECE) Semester II

Course Title : Software Development Fundamentals - II Maximum Time : 1 Hour

Course Code : 15B11CI211 Maximum Marks : 20

Course Outcomes:

CO1	Explain various object-oriented concepts like class and objects, constructors, abstraction, data hiding (access specifiers) etc.	Understand level (Level 2)
CO2	Apply object-oriented concepts like inheritance, polymorphism and templates to various real-world problems.	Apply level (Level 3)
CO3	Apply SQL commands to create tables and perform various operations like insert, delete, select on tables.	Apply level (Level 3)
CO4	Analyze the source code for possible outcomes, exceptions and debugging the errors.	Analyze level (Level 4)
CO5	Design and implement the class diagrams for varied real life problems using OOPs.	Create level (Level 5)

Note: Attempt all questions:

Q1. Design a C++ program that computes the annual tax of an employee. Assume, there is an Employee class having data members eid(int), eName(string) and annualSalary(double) and a TaxDeductibles class having data members eid(int), insurancePremium(double) and homeLoanInterest(double). Design a non-member function that can accept one object from both the classes and compute tax payable by an employee (eid should match with both the Employee and TaxDeductibles object) as per the following rules:

- A) The total deduction = insurance premium + home loan interest (Case 1).
B) The maximum limit of insurance premium is 50,000. If any employee has insurance premium above 50,000 will be considered for deduction (Case 2).
C) Annual tax = 10% of (annual salary- total deductions).

friend funcⁿ

[CO2 (Apply), 5 Marks]

Test Cases:

1. Enter Employee Details... eid:123 eName: Rohit annualSalary: 1000000 Enter Tax Deductibles... eid: 123 insurancePremium: 40000 homeLoanInterest: 160000 Your Total deduction = 40000+160000 = 200000. Your Annual Tax =10% of (1000000-200000) = 80000.	2. Enter Employee Details... eid:100 eName: Virat annualSalary: 1500000 Enter Tax Deductibles... eid: 100 insurancePremium: 20000 homeLoanInterest: 250000 Your Total deduction = 50000+250000 = 300000. Your Annual Tax =10% of (1500000-300000) = 120000.	3. Enter Employee Details... eid:123 eName: Rohit annualSalary: 1000000 Enter Tax Deductibles... eid: 324 insurancePremium: 40000 homeLoanInterest: 160000 Sorry!! Unable to compute tax as the eid don't match.
---	--	--

Q2. The position of an object in 2-dimensional space in vector notation is represented as $P=ax+by$ while the velocity of the object is represented as $V=v_1x+v_2y$. Assuming that the initial position of a boat in a lake is $(ax+by)m$ from its starting point and its velocity is $(v_1x+v_2y) m/s$, the final position of the boat after t secs (t is a scalar) can be computed as $(ax+by)+(v_1x+v_2y)*t$ from its initial position. Design a class whose objects can store the position and velocity of an object in 2D space in the given format and overload the $+$ and $*$ operator to get the final position. The initial position, velocity and the time interval must be entered by the user.

Operator overloading [CO2 (Apply), 4 Marks]

Q3. Create a Product class having data members as costPrice(double), markedPrice(double) and discount(double). A product is called a feasible product if and only if its selling price (marked price*(1-discount/100)) is greater than its cost price. Write a C++ program that will create any number of products dynamically. The program will terminate as soon as a product is created whose selling price falls below its cost price. All the products must be initialized via a parameterized constructor. Also display the number of feasible products created at the end.

Test Cases:

Enter product details: 1000 2000 15 Cost price=1000, Marked price=2000, Discount=15%, Selling Price=1700	Enter product details: 5000 7000 10 Cost price=5000, Marked price=7000, Discount=10%, Selling Price=6300	Enter product details: 1000 1200 20 A total of 2 feasible products were created.
---	---	--

[CO1 (Understand), 5 Marks]

Q4. Find the output with suitable explanation for the below given snippets:

Output - Baulh

<pre>#include<iostream> using namespace std; class Number{ public: int x; Number(){ x=10; } Number(int y=5){ x=y; } void display(){ cout<<x; } }; int main(){ Number obj1; obj1.display(); return 0; }</pre>	<pre>#include<iostream> void modifyArray(int *arr, int size){ for(int i=0;i<size;++i){ *arr = (*arr++ * 2) + 5; arr++; } } int main(){ int arr[]={10,20,30,40,50}; int len=sizeof(arr)/sizeof(arr[0]); modifyArray(arr,len); std::cout<<"Modified Array:"; for(int i=0;i<len;i++) std::cout<<arr[i]<<" "; return 0; }</pre>	<pre>#include<iostream> void processArray(int* arr, int size){ for(int i=0;i<size;++i){ arr[i] = 2*arr[i] * 2; } } int main(){ int * originalArray=new int[5]; for(int i=0;i<5;++i){ originalArray[i] = i*i + 1; } processArray(originalArray,5); for(int i=0;i<5;++i){ std::cout<<originalArray[i]<<" "; } delete[] originalArray; return 0; }</pre>
--	--	---

[CO1 (Understand), 6 Marks]